C. Reverse XOR

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: bitmasks | https://codeforces.com/problemset/problem/2160/C

Given a **positive** integer x, let $f(x)$ be the positive integer formed by reversing the binary representation of x without leading zeroes. For example, if $x=12=1100_2$, then $f(x)=0011_2=3$.

You are given an integer n. Please determine if there exists a positive integer x such that $x \oplus f(x) = n$[1].

[1]Here, $\oplus$ denotes the bitwise XOR operation.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($0 \le n$ lt 2^{30}).

Output

For each test case, output YES if there exists a positive integer x such that $x \oplus f(x) = n$, and NO otherwise.

You can output the answer in any case. For example, the strings "yEs", "yes", and "Yes" are also recognized as positive responses.

Example

Input
6 0 3 6 8 10 11

Output
YES YES YES NO YES NO

Note

In the first case, when $x=1$, $f(x)=1$, and $x \oplus f(x)=0$. Thus, the answer is YES.

In the second case, when $x=2$, $f(x)=1$, and $x \oplus f(x)=3$. Thus, the answer is YES.

In the fourth test case, we can show there is no x that satisfies $x \oplus f(x)=8$, so the answer is NO.